

Speedup over PyTorch (1 GPU)

Strong Scaling

Weak Scaling

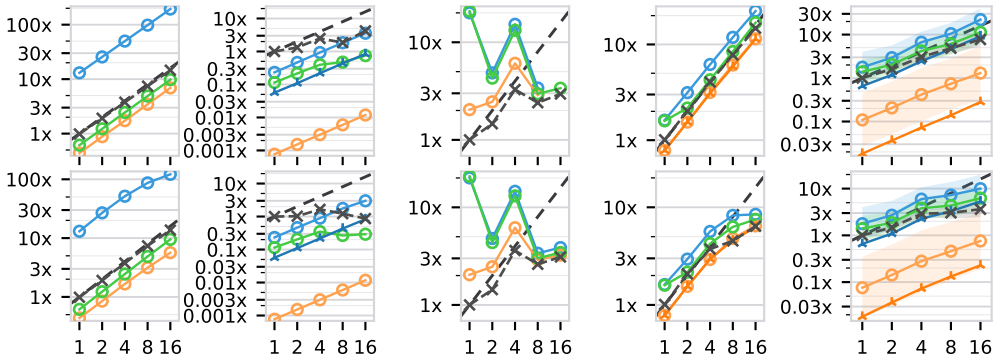
MoE Router

SDPA

All-to-All

Vocab Softmax

Geomean

GPUs P

- Ideal Scaling of the PyTorch Baseline
- ⊗ PyTorch Distributed ($n=19$)
- Qwen3.8-27B, No Skill Packet ($n=15$)
- ⊕ Qwen3.8-27B, RCCL Hints ($n=5$)
- GPT-OSS-120B, No Skill Packet ($n=14$)
- ⊕ GPT-OSS-120B, RCCL Hints ($n=3$)
- Kimi-K2.7-Code, No Skill Packet ($n=16$)